# CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

Generated on 2026-10-25 13:08

## Run

| | |
|---|---|
| **Split** | train |
| **Stimulus penalisation** | 1.0 |
| **Prior penalisation** | 1.0 |
| **Genes** | 100 |
| **Stimuli** | 2 |
| **Observed cells** | 5413 |
| **Timepoints** | 0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432 |
| **Cell types** | Neural, MEF, Epithelial, Stromal, Trophoblast, iPSC |
| **Network shown** | inter_simul.npy |
| **Embeddings** | UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic) |

## Perturbations (perturbation_simulation)

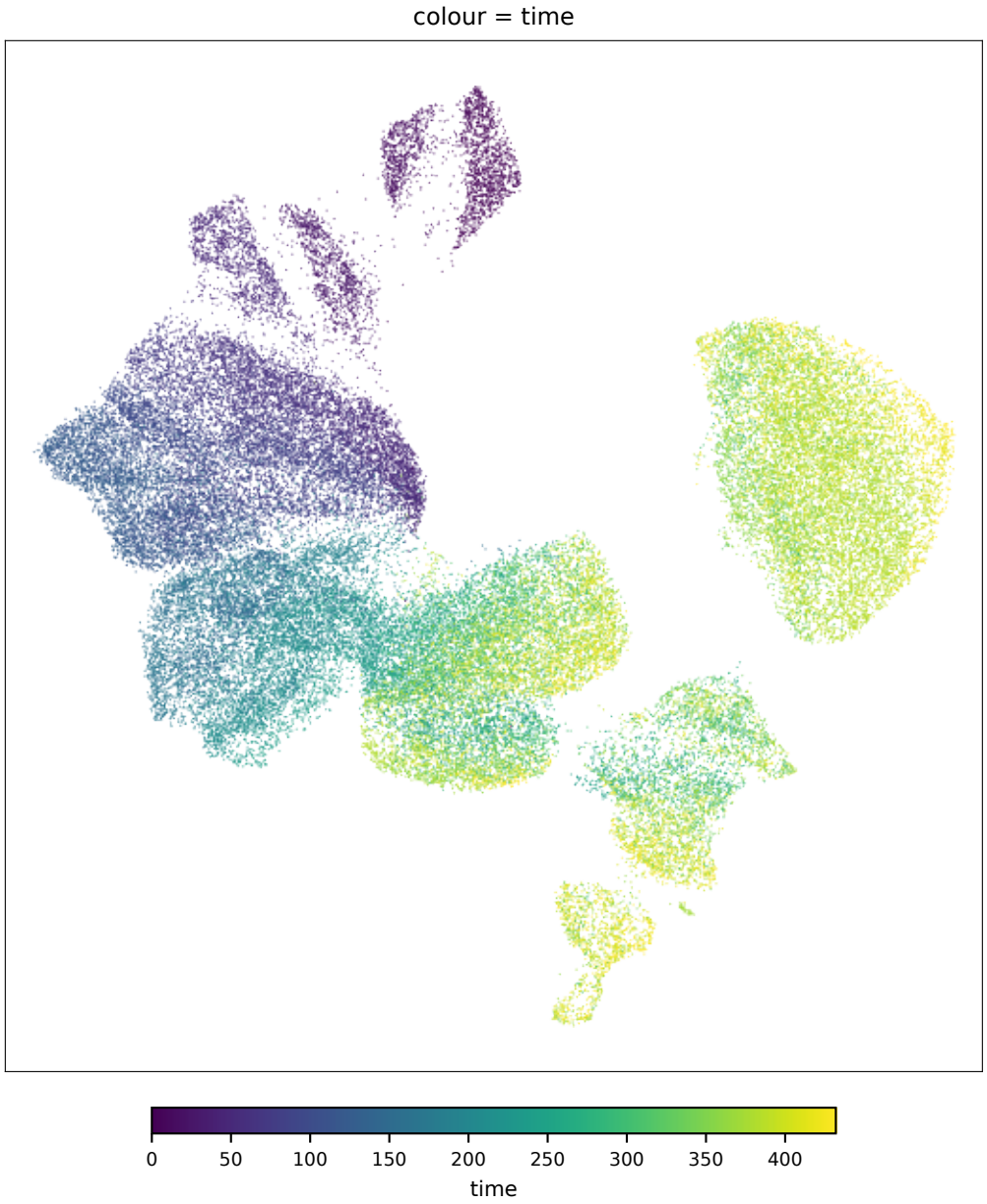
- ✓ P1 OV Obox6
- ✓ P2 OV Zfp42

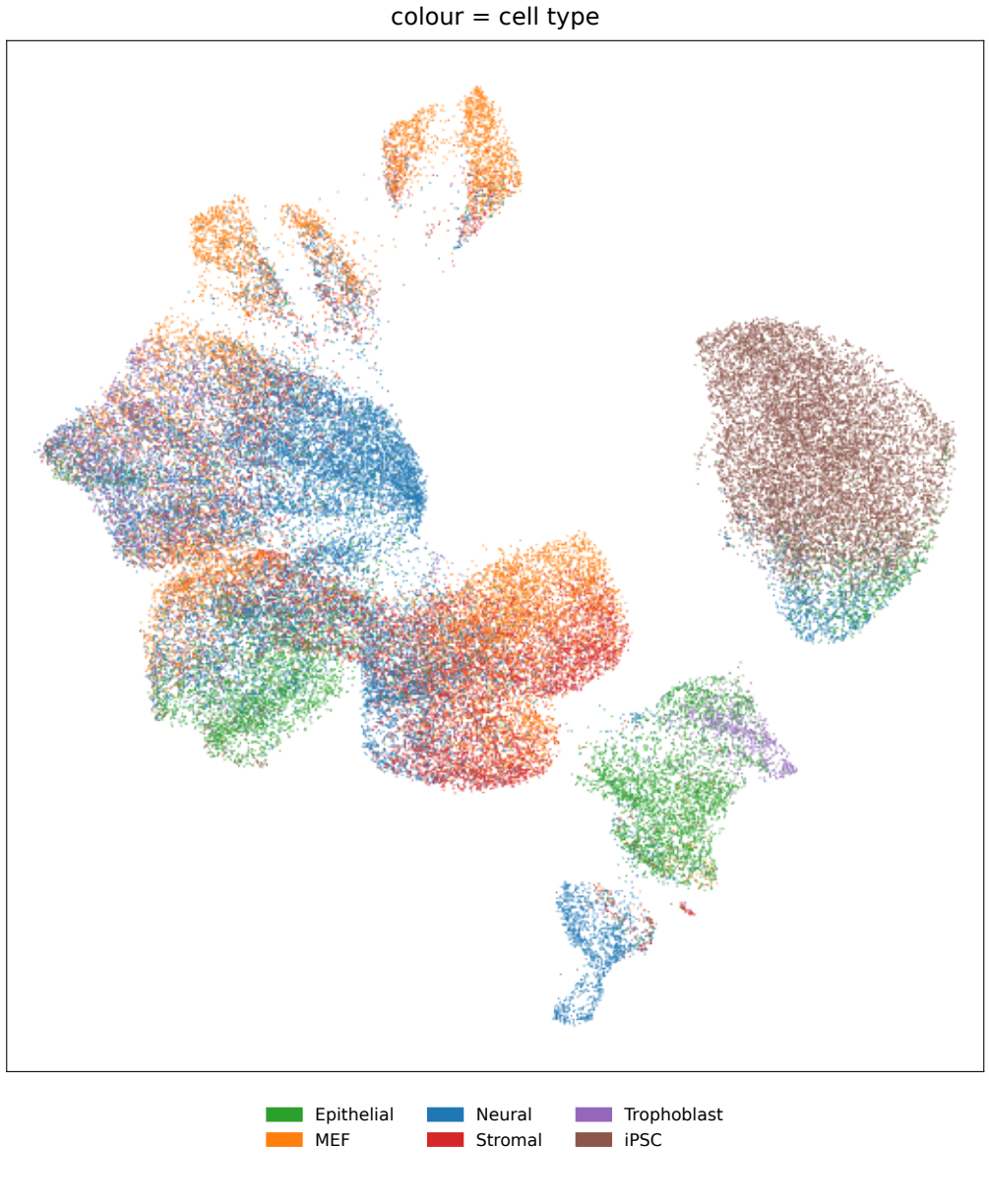
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2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

# Data — UMAP of all the cells

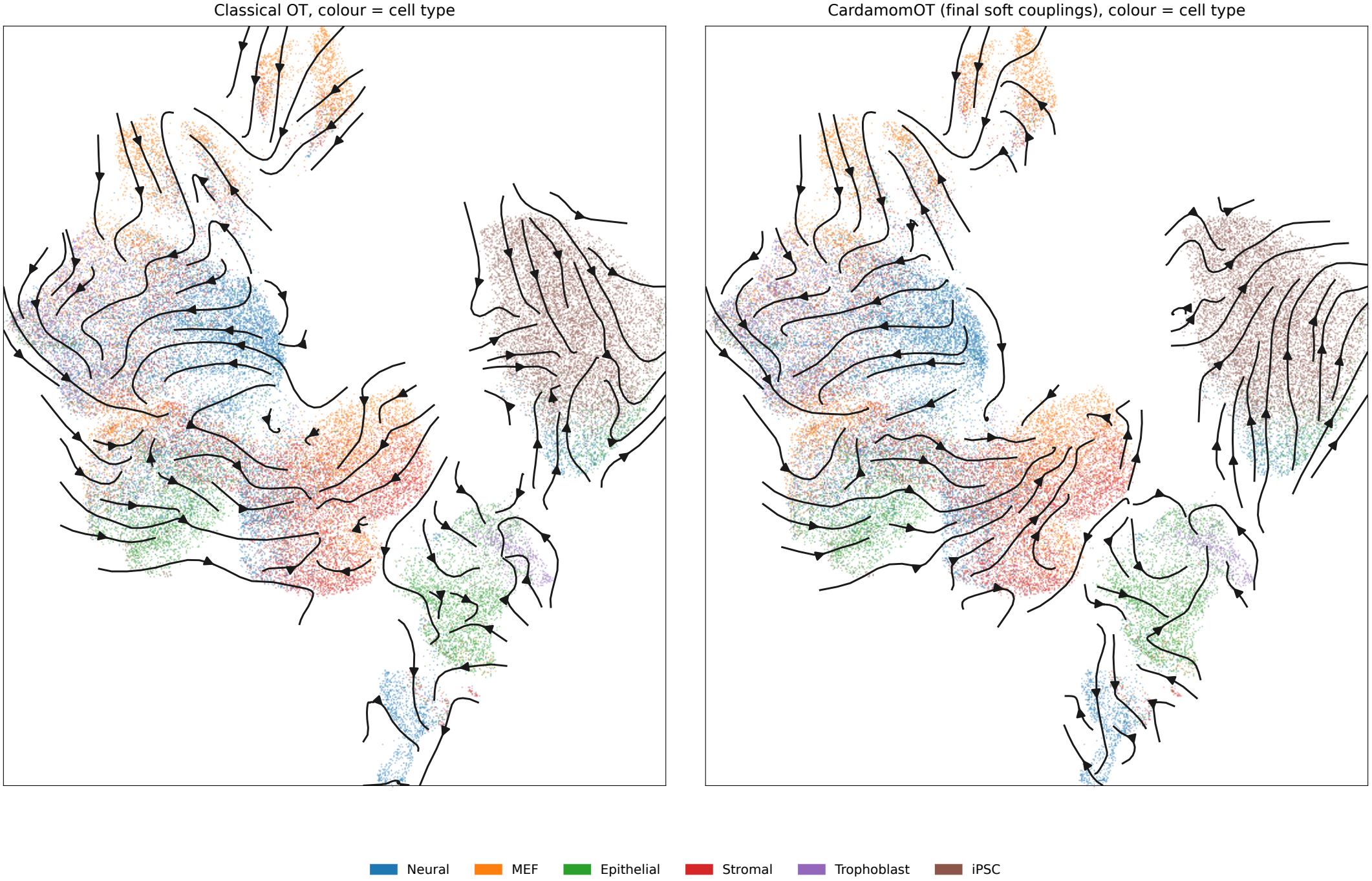
Data/data.h5ad, 50897 cells: log1p(counts), 2000 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.



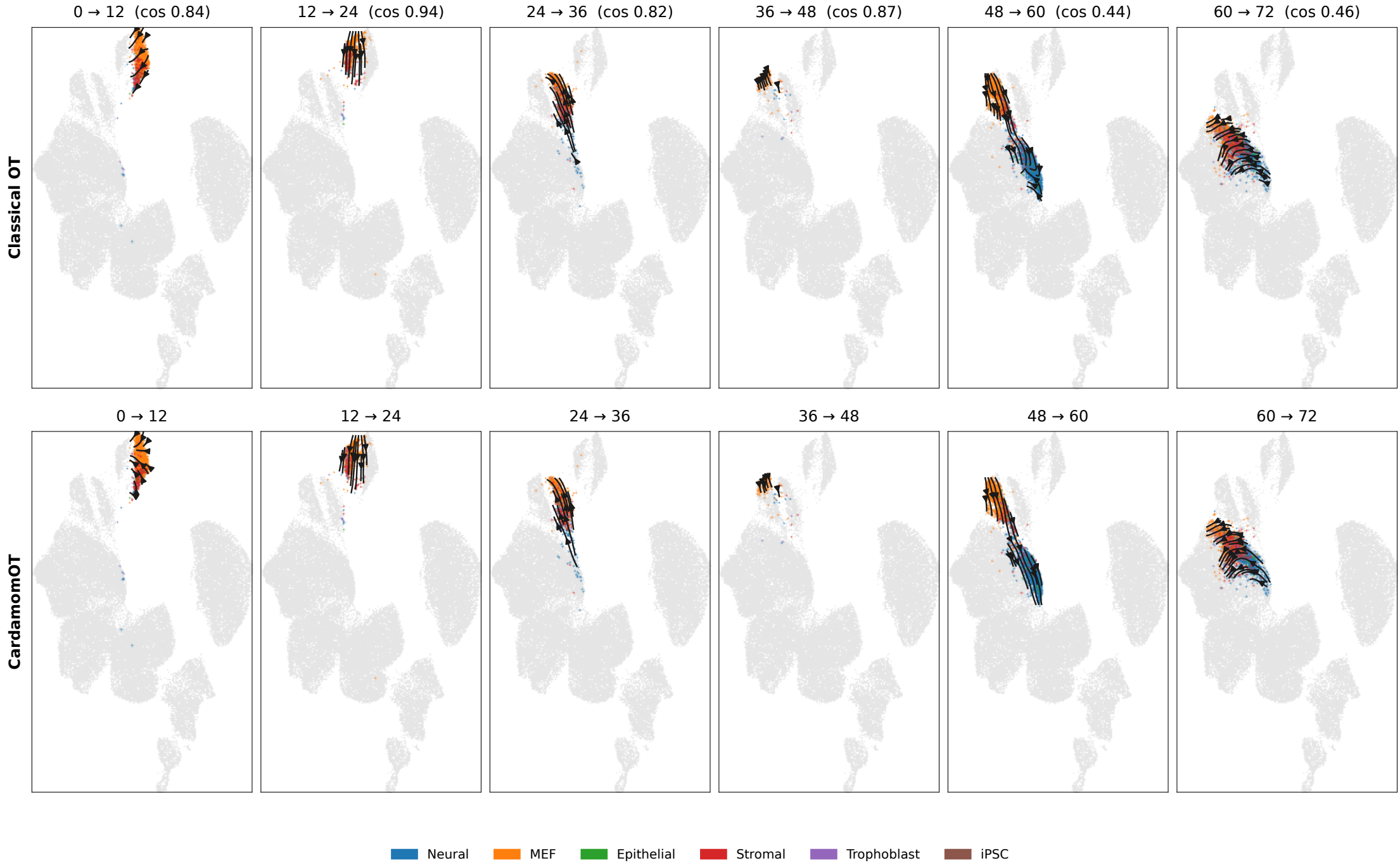
# Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.62 (PCA space). CardamomOT: final soft couplings of the trajectories (5413 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



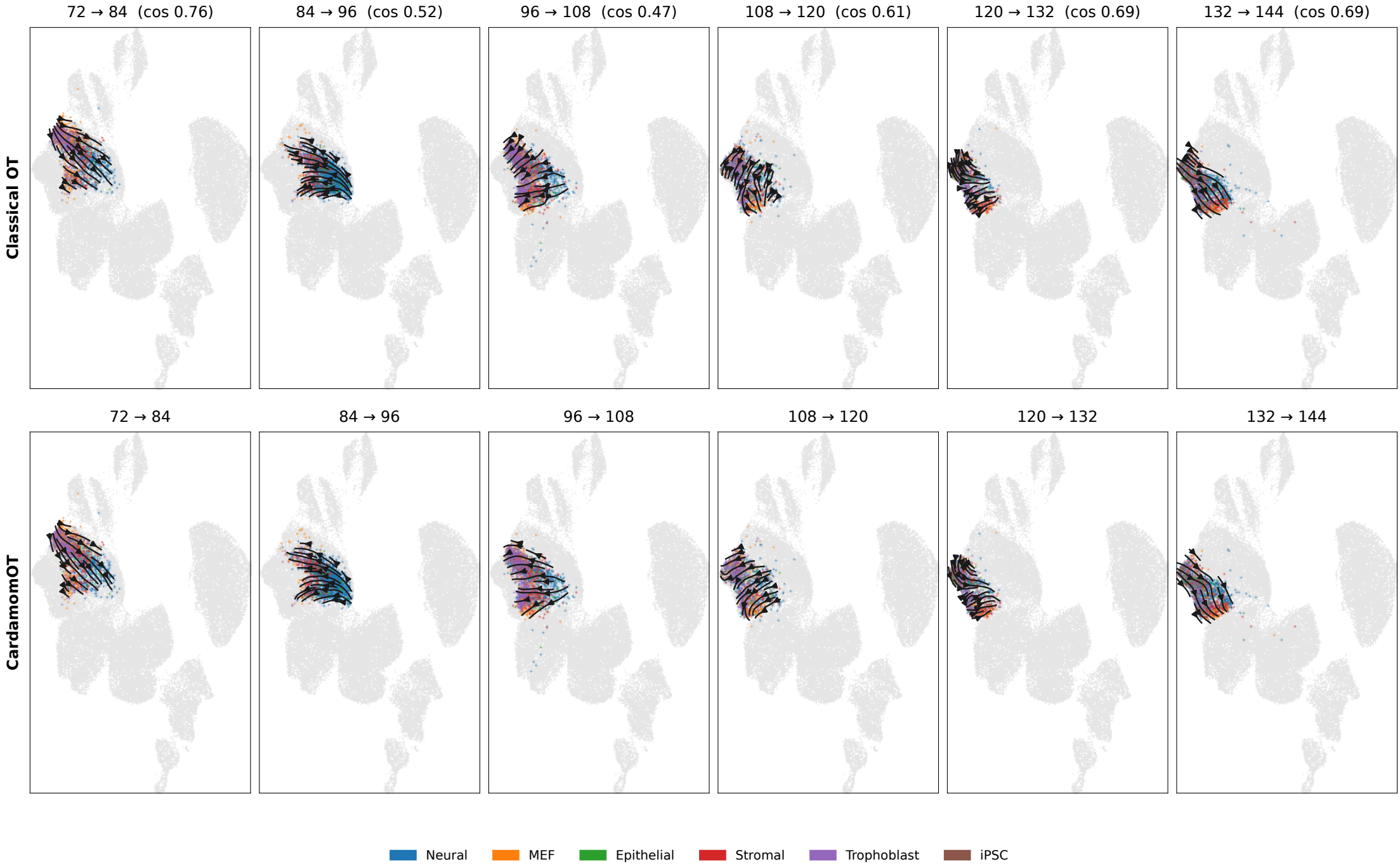
# Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



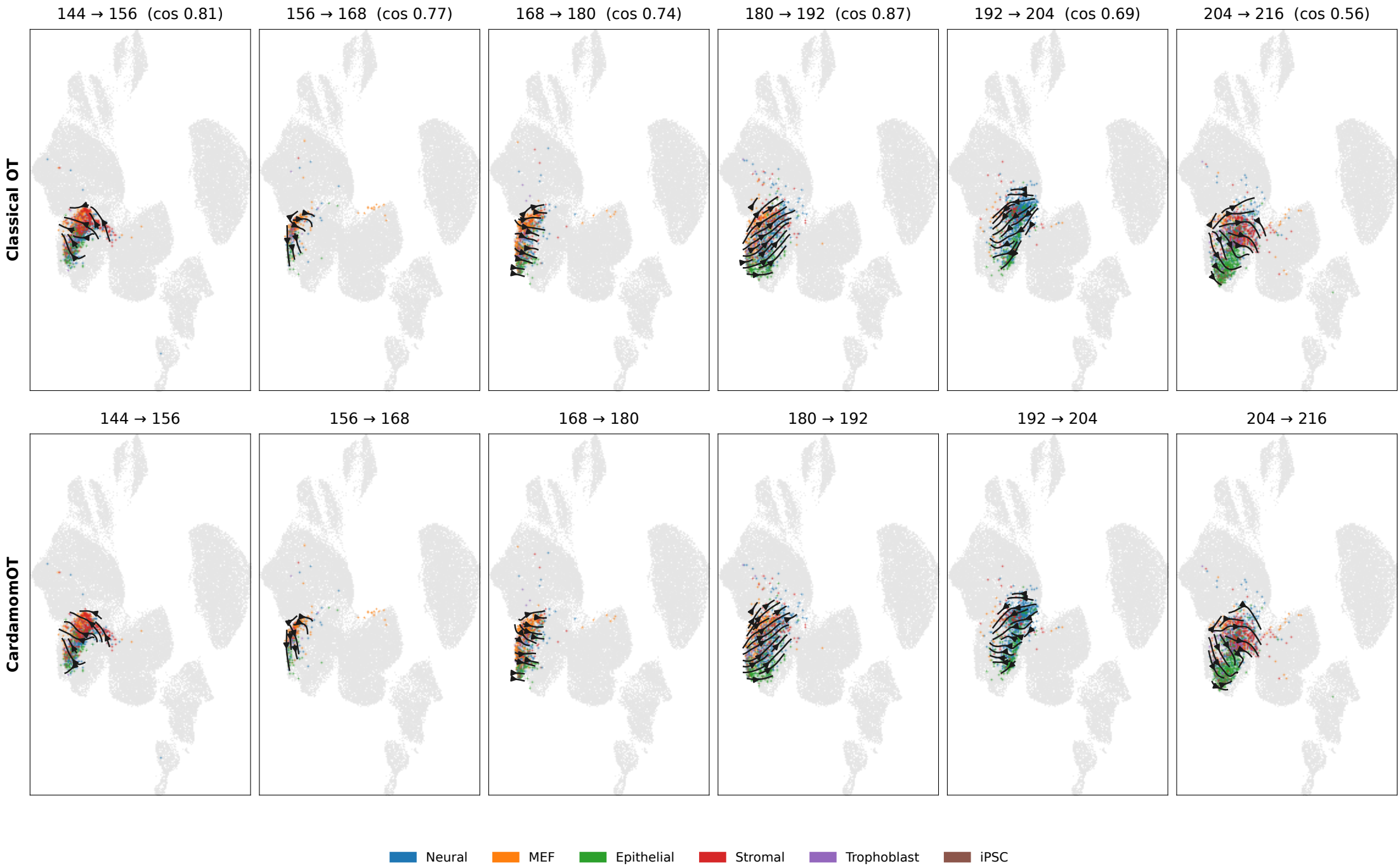
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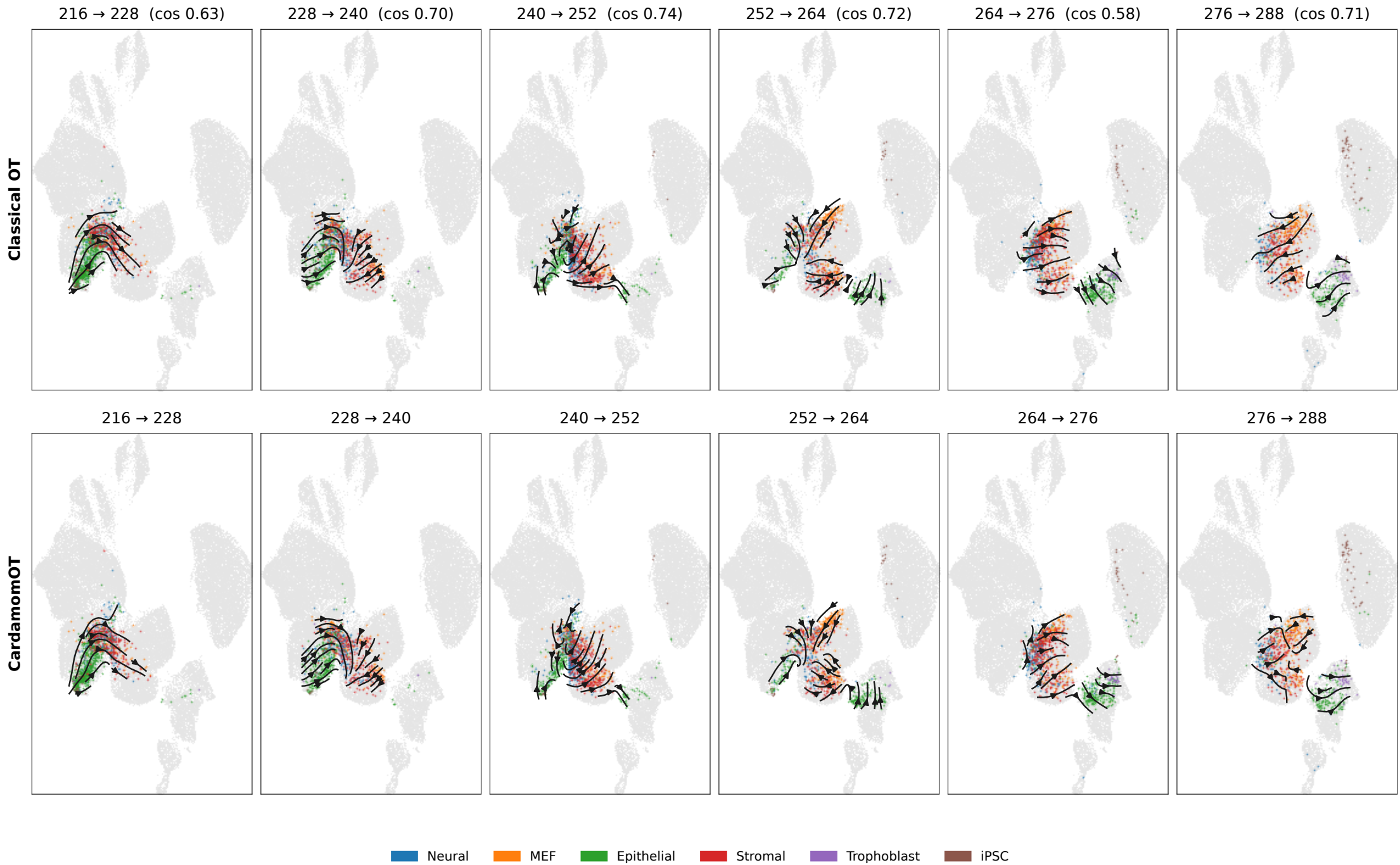
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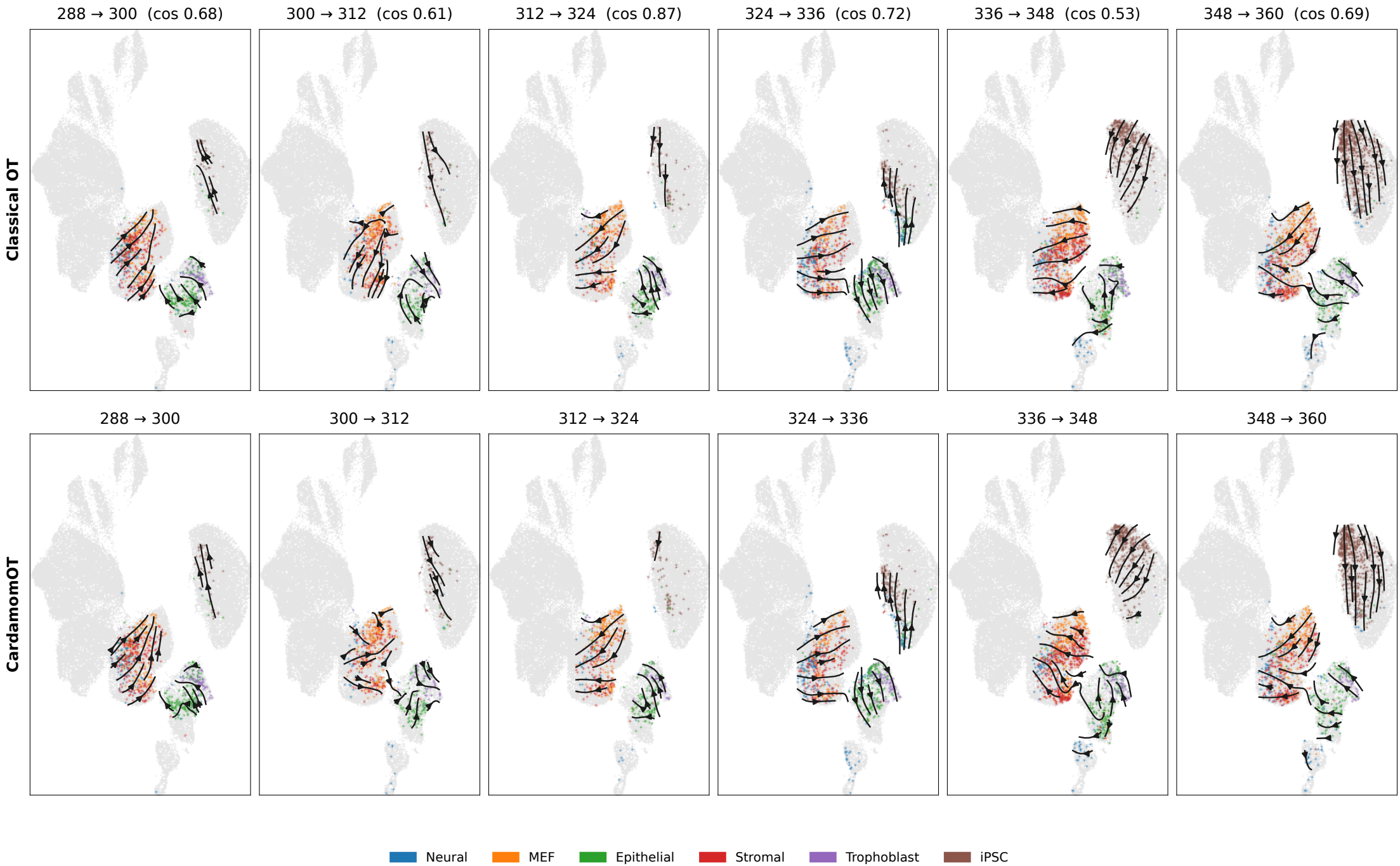
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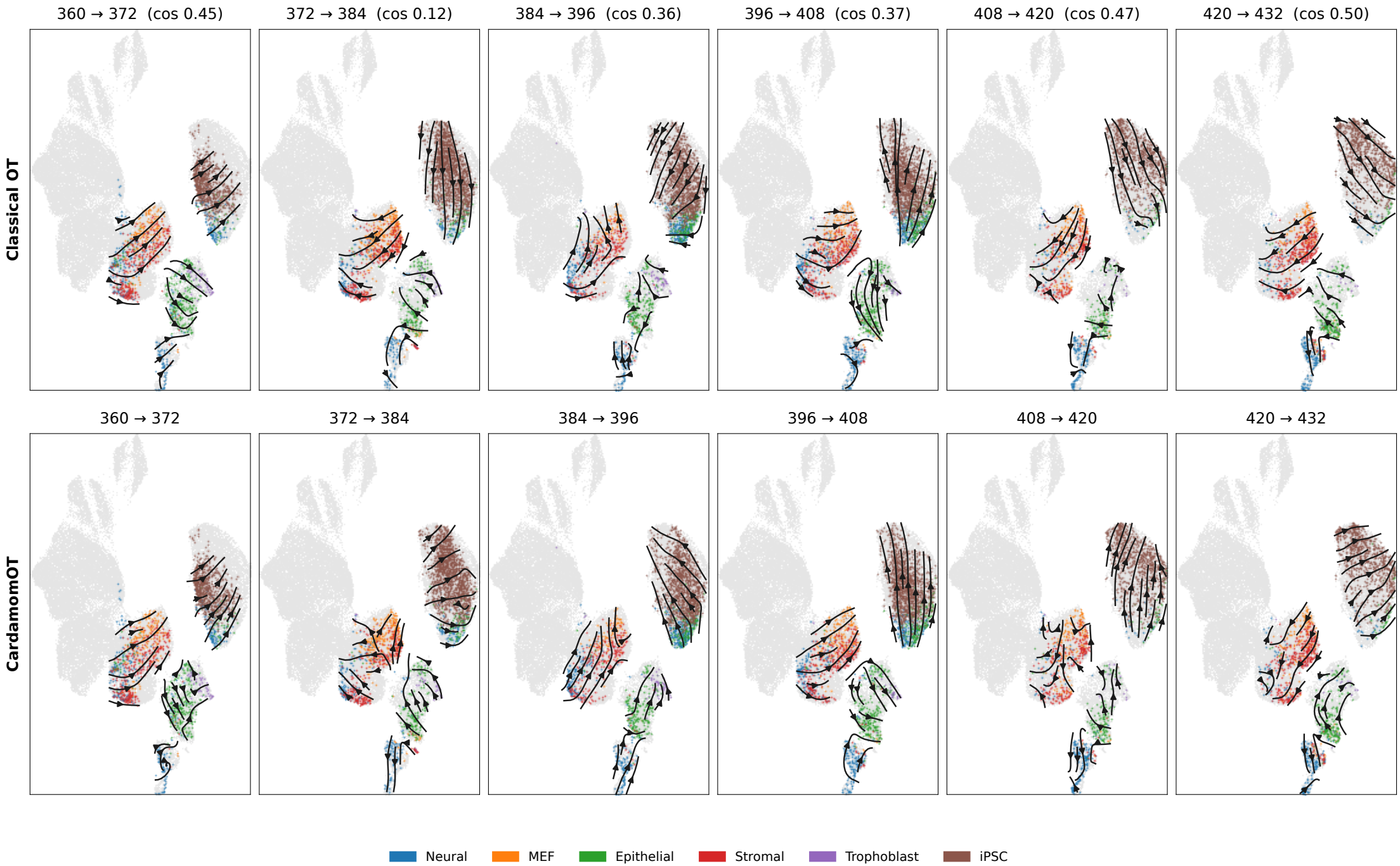
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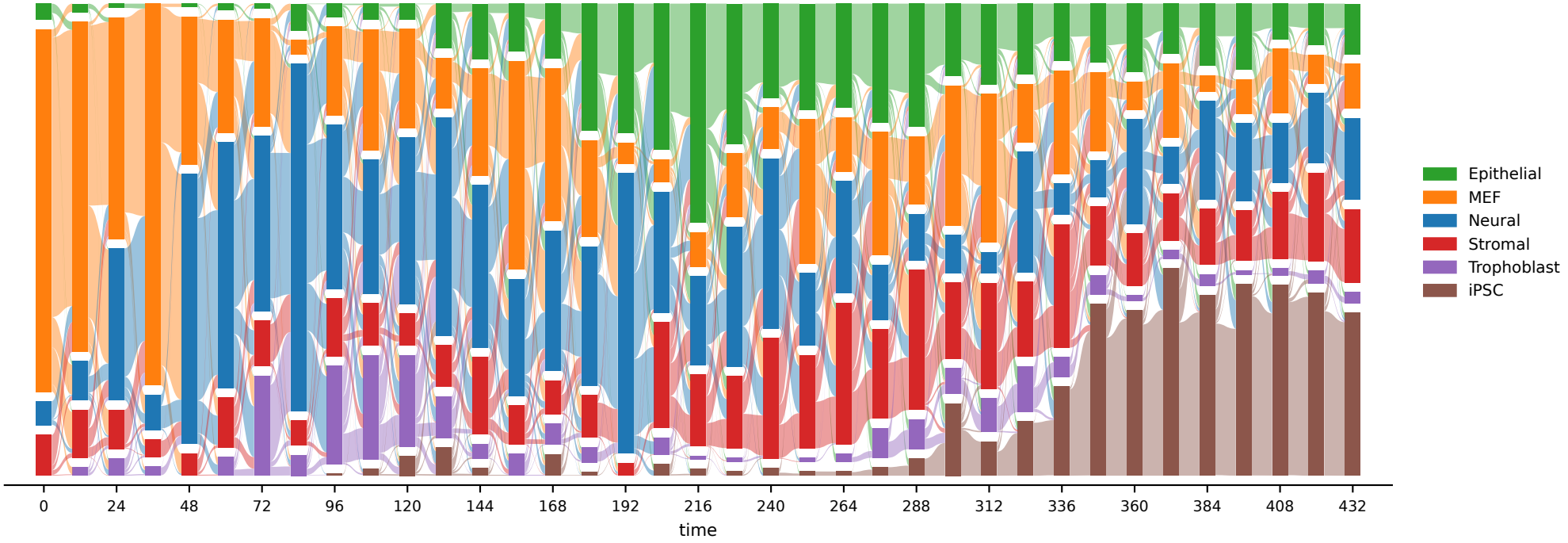
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Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



# Classical OT — cell-type transitions and fate genes

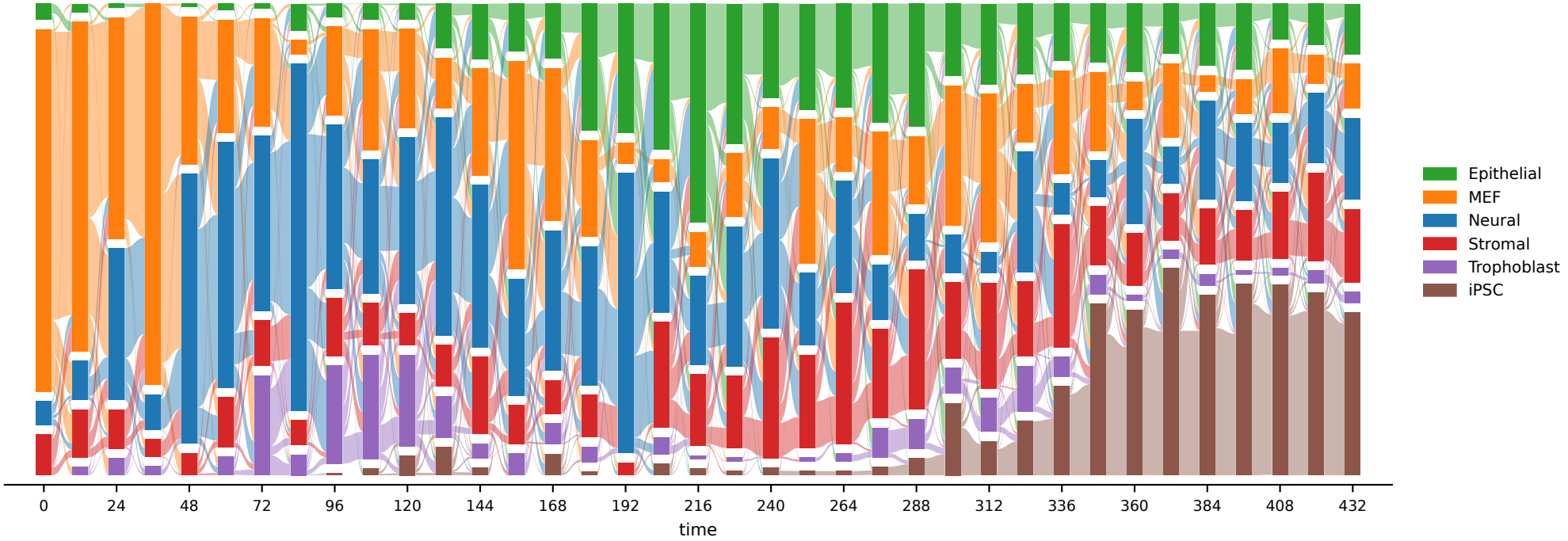
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).




| | Top-10 genes up in the ancestors (vs ancestors of the other cells) | Top-10 genes up in the cell type (vs the other cells) |
|---|---|---|
| Epithelial | **Krt8** **Krt18** **Epcam** **Krt7** **Dbi** **Bbx** **Mbnl3** Marcksl1 **Xist** **S100a14** | **Krt8** **Krt18** **Epcam** **Krt7** **Bbx** **Mbnl3** **S100a14** **Dbi** **Xist** Hcfc1r1 |
| MEF | **Col1a2** **Thbs1** **Col1a1** **Tagln** **Vim** **Timp2** **Timp3** Sparc **Cd44** Emp3 | **Col1a2** **Vim** **Tagln** **Thbs1** **Timp2** **Cd44** Tpm1 **Col1a1** **Timp3** Fbn1 |
| Neural | **Vim** **Ccnd2** **S100a10** **Xist** **Cks1b** **Emp3** **Id3** **Mras** **Pcolce2** Myc | **Vim** **Ccnd2** **S100a10** **Xist** **Id3** **Cks1b** **Pcolce2** **Mras** Cenpa **Emp3** |
| Stromal | **Mgp** **Sparc** **Col1a2** **Col1a1** **Vim** **Timp2** **Emp3** **Selm** Id3 **Laptm4a** | **Sparc** **Mgp** **Col1a2** **Col1a1** **Vim** **Emp3** **Timp2** **Selm** **Laptm4a** Efemp2 |
| Trophoblast | **Sri** **Dbi** **Hsp90aa1** **Hmgb2** Krt8 Krt18 **Rbbp7** Mbnl3 Cks1b **Hmgn5** | **Dbi** **Sri** **Hsp90aa1** S100a10 **Rbbp7** **Hmgn5** **Hmgb2** Cyb5r3 Taldo1 Cdkn1c |
| iPSC | **Dppa5a** **Hsp90aa1** **Mybl2** **Ooep** **Nanog** **Nfatc2ip** **Cdc5l** **Sox2** **Gjb3** Zfp42 | **Dppa5a** **Hsp90aa1** **Ooep** **Mybl2** **Nanog** **Cdc5l** **Nfatc2ip** **Sox2** **Gjb3** Fmr1nb |

# Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).




| | Top-10 genes up in the ancestors (vs ancestors of the other cells) | Top-10 genes up in the cell type (vs the other cells) |
|---|---|---|
| **Epithelial** | **Krt8** **Epcam** **Krt18** **Krt7** **S100a14** **Bbx** **Mbnl3** Dbi Dmrtc2 **Gpx2** | **Krt8** **Epcam** **Krt18** **Krt7** **Mbnl3** **S100a14** **Bbx** Cdkn1c Gpx3 **Gpx2** |
| **MEF** | **Col1a2** **Vim** **Emp3** **Thbs1** **Tpm1** Col1a1 Sparc **Timp2** **Cd44** **Tagln** | **Vim** **Tpm1** **Col1a2** Ccnd2 **Cd44** **Emp3** **Tagln** **Thbs1** S100a10 **Timp2** |
| **Neural** | **Ccnd2** **Vim** **Dmrtc2** **Myc** **Pcolce2** **Cenpa** **Mras** **Cks1b** S100a10 Sms | **Dmrtc2** **Vim** **Ccnd2** Id3 **Pcolce2** **Mras** **Cenpa** **Myc** **Cks1b** Crabp2 |
| **Stromal** | **Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** Id3 **Timp2** Prrx1 Cd81 **Selm** | **Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** Emp3 **Selm** Xist Laptm4a **Timp2** |
| **Trophoblast** | **Sri** **Hsp90aa1** **Hmgb2** **Dbi** **Cks1b** Krt18 Mbnl3 Krt8 **Rbbp7** Cenpa | **Dbi** **Sri** **Hsp90aa1** S100a10 **Rbbp7** **Hmgb2** Hmgn5 Taldo1 **Cks1b** Cyb5r3 |
| **iPSC** | **Dppa5a** **Hsp90aa1** **Mybl2** **Nanog** **Ooep** **Sox2** **Nfatc2ip** Gjb3 **Zfp42** **Cdc5l** | **Dppa5a** **Hsp90aa1** **Ooep** **Mybl2** **Nanog** **Cdc5l** **Nfatc2ip** **Sox2** Fmr1nb **Zfp42** |

# Genes of the model (100)

Data/data_train.h5ad, alphabetical order, coloured by selection role. Italics: no edge inside the selection (probability ≥ selection_edge_prob, literature-feasible;
0 genes). Trajectory preservation (classical OT on these genes vs every gene; network otvelo_granger): velocity cosine 0.88 (random genes 0.77), fate JS distance …

| | | | |
|---|---|---|---|
| Antxr1 | Efemp2 | Ltbr | S100a10 |
| Anxa3 | Ehd2 | Marcksl1 | S100a14 |
| Batf3 | Emp3 | Mbnl3 | Sall4 |
| Bbx | Epcam | Mgp | Sat1 |
| Bhlhe40 | Eva1b | Mras | Sdc4 |
| Btg2 | Fbn1 | Msmo1 | Selm |
| Ccnb1 | Fmr1nb | Mybl2 | Sgol1 |
| Ccnd2 | Foxo4 | Myc | Sms |
| Cd44 | Gata2 | Nanog | Sox2 |
| Cd81 | Gjb3 | Nasp | Sparc |
| Cdc5l | Gnl3 | Nfatc2ip | Sri |
| Cdkn1c | Gpx2 | Nfic | Tagln |
| Cenpa | Gpx3 | Nfix | Taldo1 |
| Cited2 | Hcfc1r1 | Obox6 | Tcl1 |
| Cks1b | Hmgb2 | Ooep | Tgif1 |
| Col1a1 | Hmgn5 | Pcolce2 | Thbs1 |
| Col1a2 | Hsp90aa1 | Phlda2 | Timp2 |
| Crabp2 | Id3 | Prnp | Timp3 |
| Cyb5r3 | Ier3 | Prrx1 | Tmbim1 |
| Dbi | Igfbp3 | Pttg1ip | Tnc |
| Ddit3 | Klf4 | Rbbp7 | Tpm1 |
| Ddx21 | Krt18 | Rhoc | Tspo |
| Dmrtc2 | Krt7 | Rhox6 | Vim |
| Dppa5a | Krt8 | Rhox9 | Xist |
| Ebna1bp2 | Laptm4a | Rrp9 | Zfp42 |

perturbed (2) query (26) driver (34) entropy (38)

# 0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) (depth model 'mean') when the depth factor s_i is used.

**Depth**

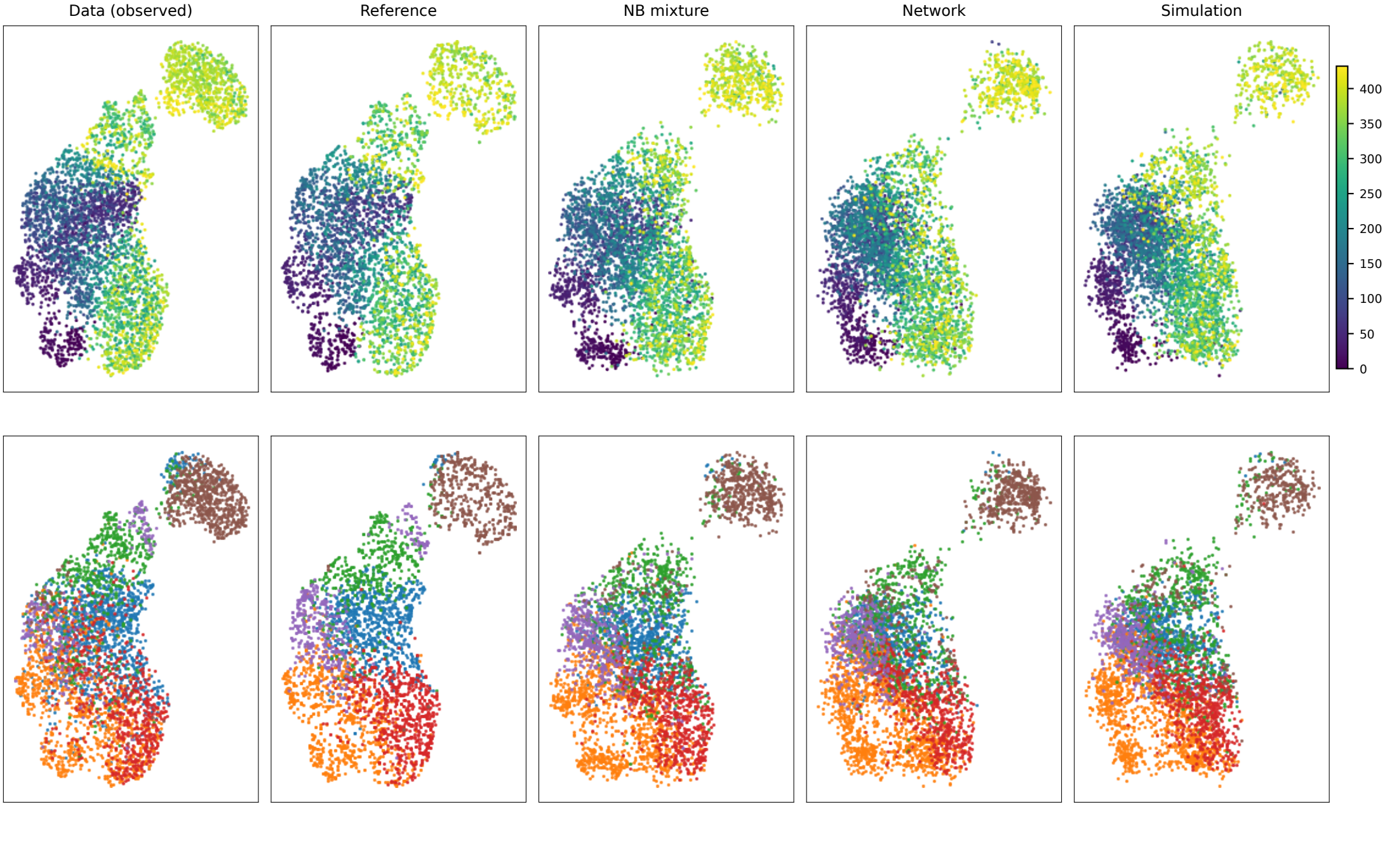
not estimated

**Reason**

100 genes < 10000

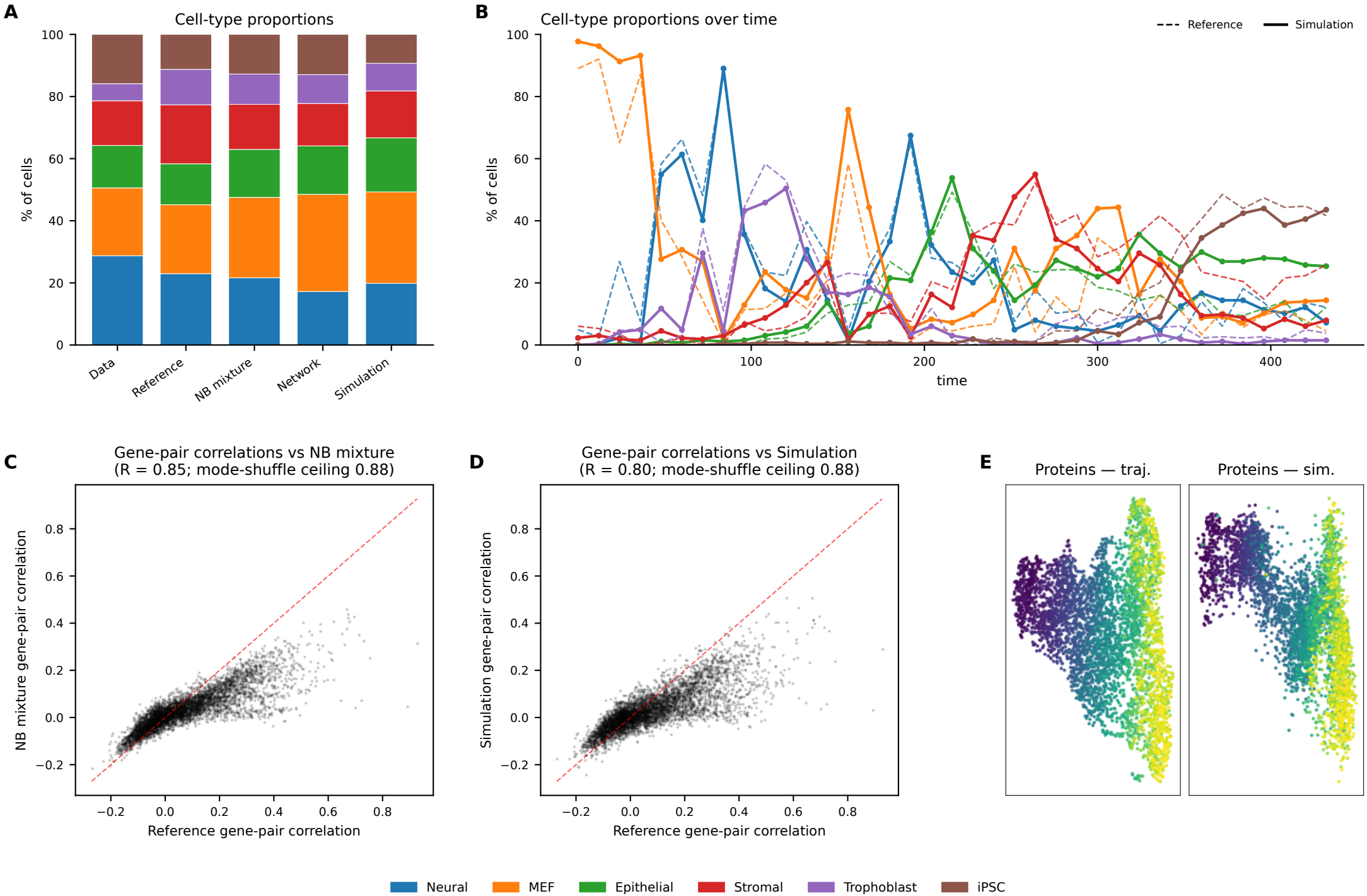
# 1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



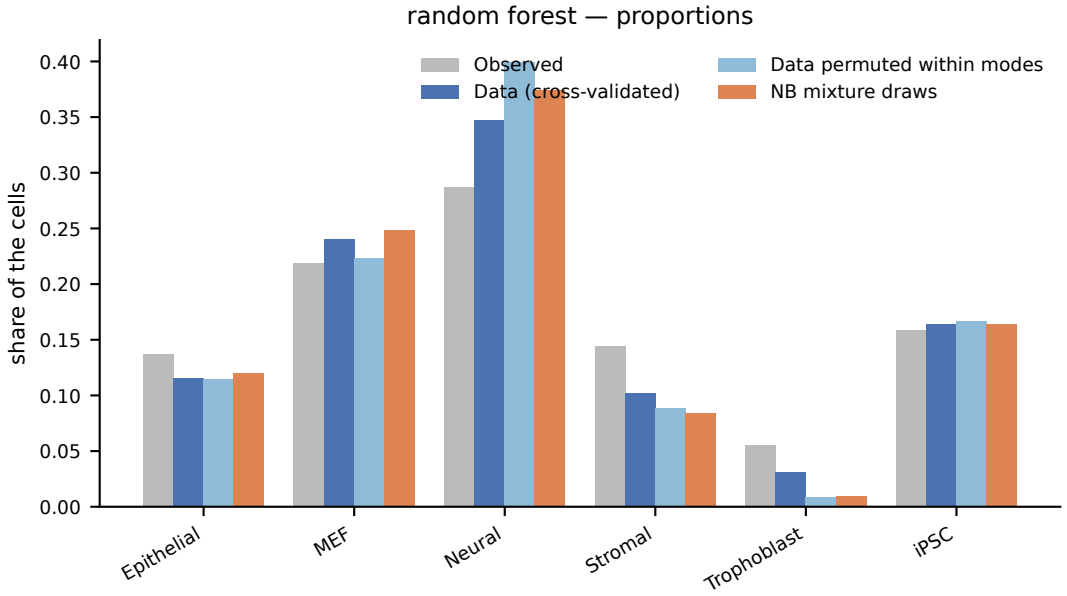
# 1. Generative model — quantitative checks

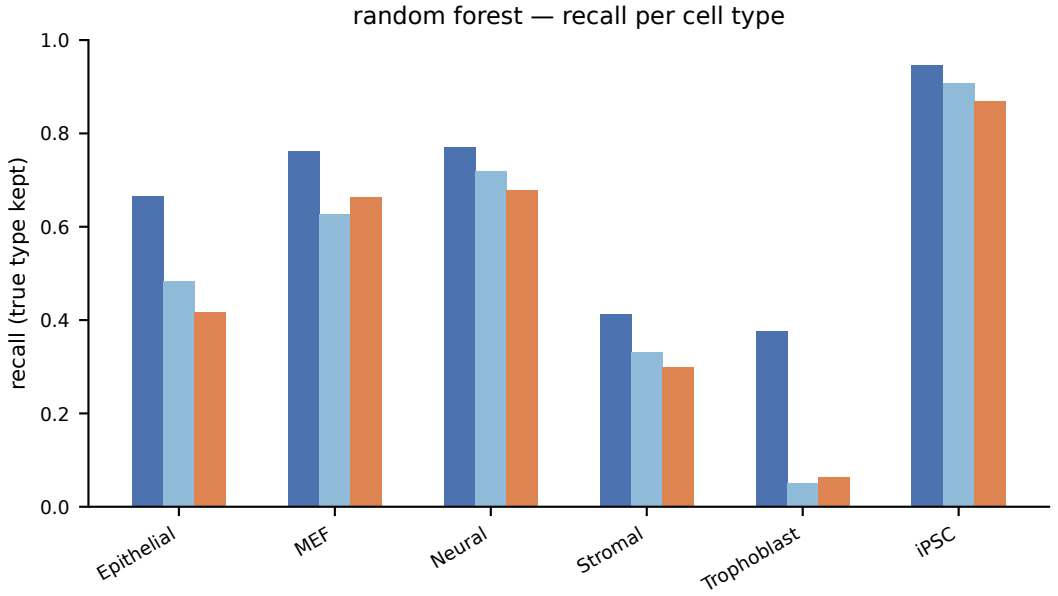
Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

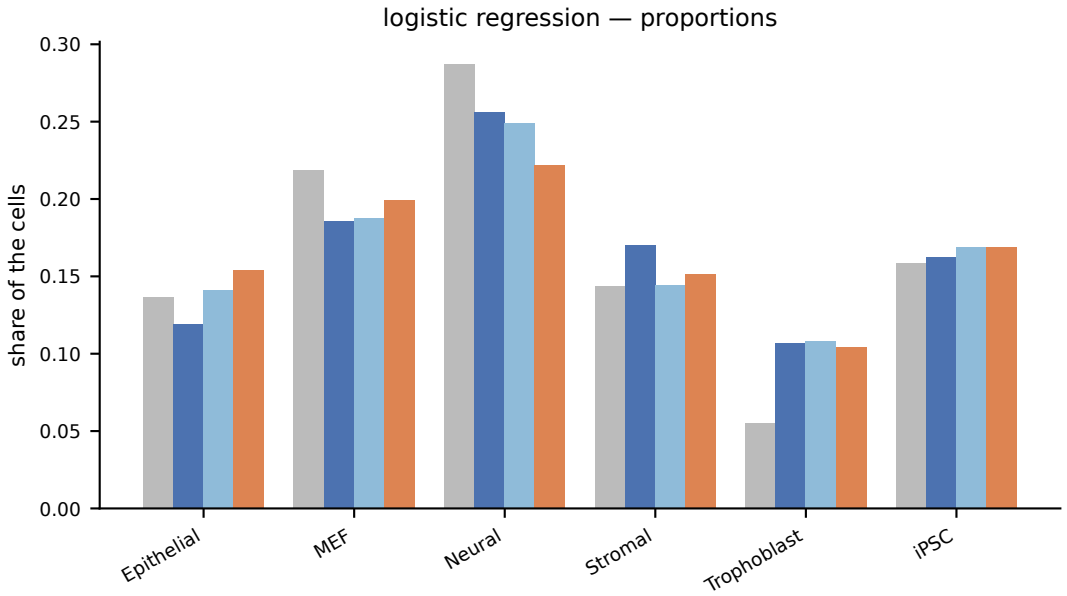


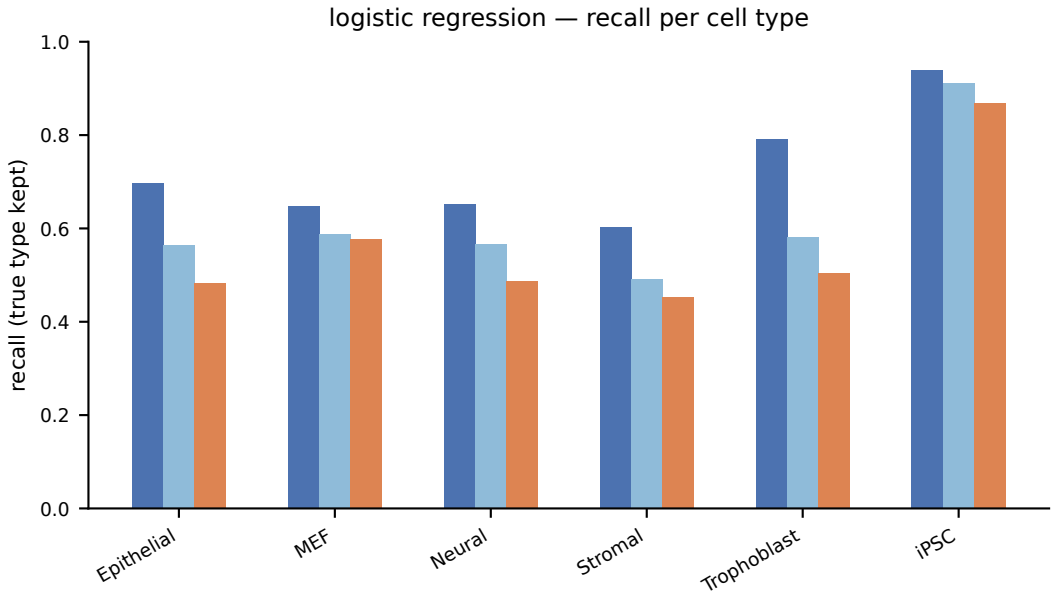
# 1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.



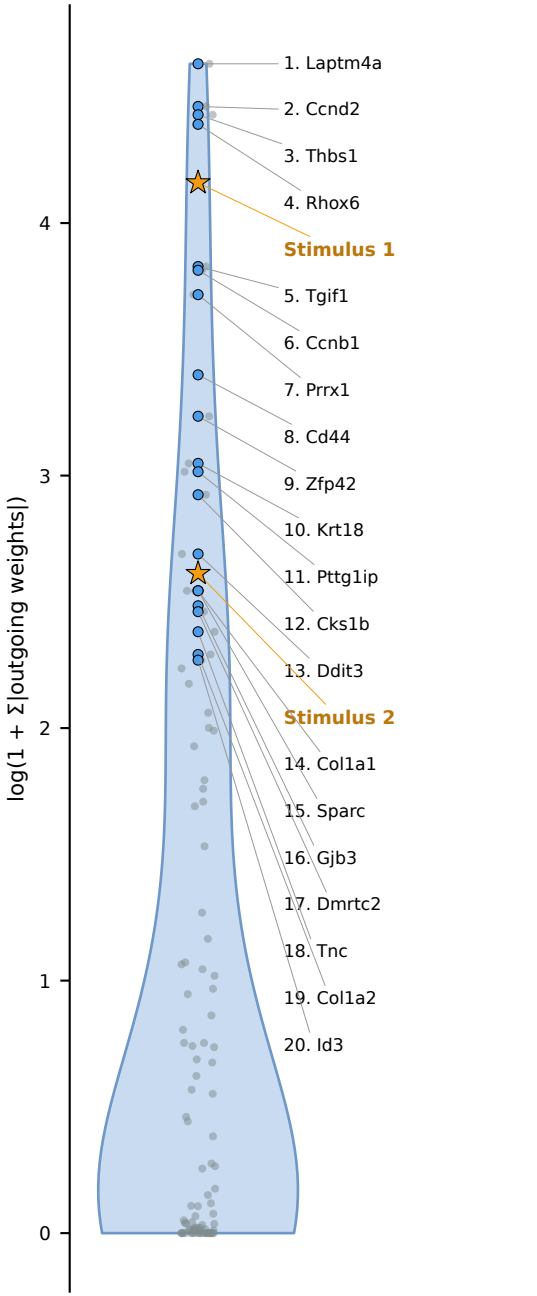



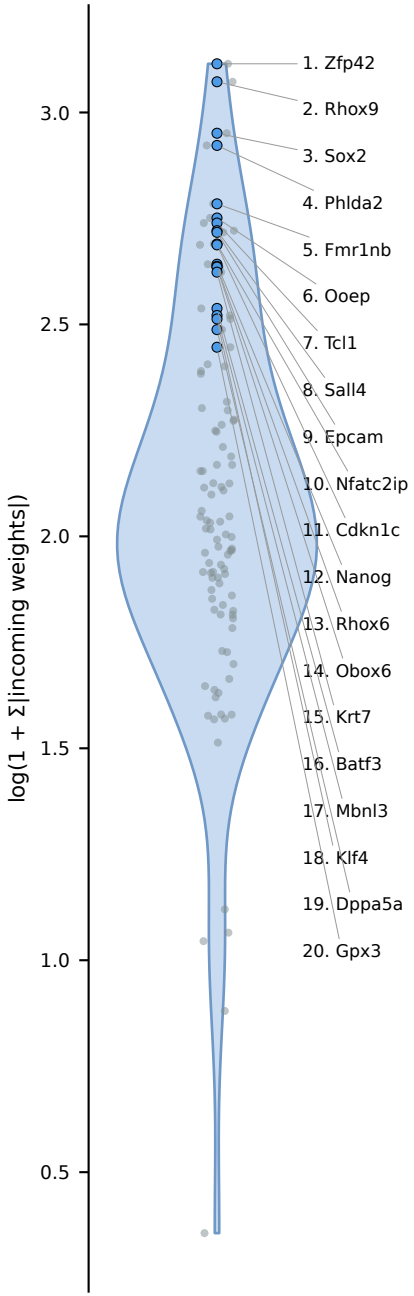
# 2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

**A**




**B**




**C** Top 20 regulators

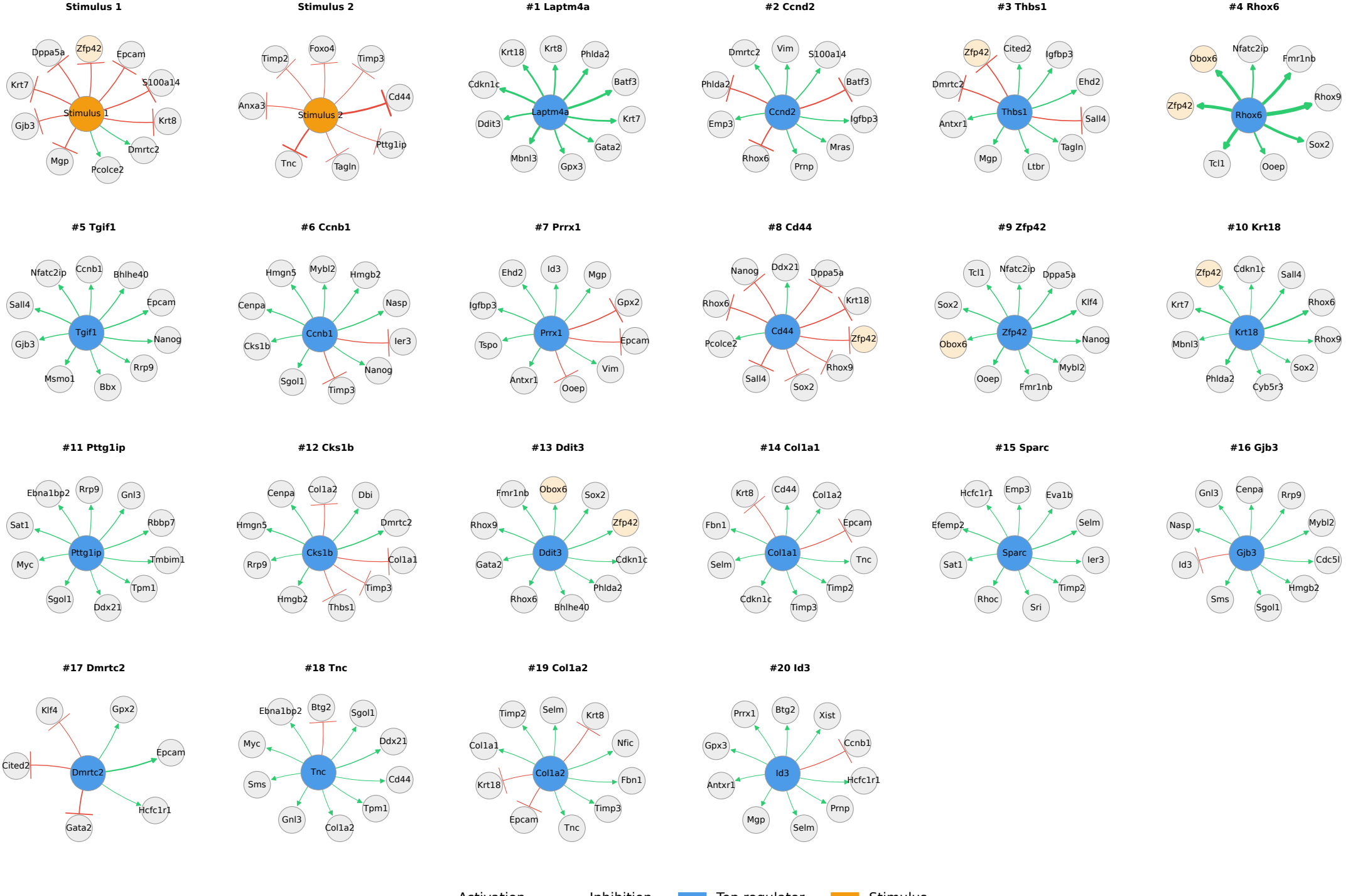
| # | Regulator | Σ\|w\| | act. | inh. |
|---|---|---|---|---|
| 1 | Laptm4a | 101.60 | 44 | 1 |
| 2 | Ccnd2 | 85.65 | 45 | 14 |
| 3 | Thbs1 | 82.85 | 47 | 11 |
| 4 | Rhox6 | 79.73 | 8 | 0 |
| 5 | Tgif1 | 44.97 | 38 | 2 |
| 6 | Ccnb1 | 44.29 | 22 | 21 |
| 7 | Prrx1 | 40.12 | 38 | 6 |
| 8 | Cd44 | 28.93 | 14 | 14 |
| 9 | Zfp42 | 24.41 | 15 | 2 |
| 10 | Krt18 | 20.08 | 14 | 1 |
| 11 | Pttg1ip | 19.40 | 19 | 1 |
| 12 | Cks1b | 17.62 | 15 | 5 |
| 13 | Ddit3 | 13.73 | 14 | 1 |
| 14 | Col1a1 | 11.73 | 20 | 2 |
| 15 | Sparc | 11.73 | 12 | 0 |
| 16 | Gjb3 | 10.99 | 12 | 2 |
| 17 | Dmrtc2 | 10.72 | 3 | 3 |
| 18 | Tnc | 9.82 | 13 | 2 |
| 19 | Col1a2 | 8.89 | 14 | 4 |
| 20 | Id3 | 8.67 | 17 | 3 |
| ★ | Stimulus 1 | 63.06 | 25 | 37 |
| ★ | Stimulus 2 | 12.63 | 0 | 8 |

**D** Top 20 regulated genes

| # | Target | Σ\|w\| | act. | inh. |
|---|---|---|---|---|
| 1 | Zfp42 | 21.53 | 3 | 4 |
| 2 | Rhox9 | 20.59 | 4 | 2 |
| 3 | Sox2 | 18.13 | 4 | 4 |
| 4 | Phlda2 | 17.58 | 7 | 2 |
| 5 | Fmr1nb | 15.19 | 4 | 0 |
| 6 | Ooep | 14.66 | 4 | 4 |
| 7 | Tcl1 | 14.47 | 3 | 0 |
| 8 | Sall4 | 14.20 | 4 | 3 |
| 9 | Epcam | 14.14 | 3 | 7 |
| 10 | Nfatc2ip | 13.73 | 8 | 3 |
| 11 | Cdkn1c | 13.70 | 6 | 2 |
| 12 | Nanog | 13.04 | 8 | 4 |
| 13 | Rhox6 | 12.97 | 4 | 2 |
| 14 | Obox6 | 12.95 | 3 | 0 |
| 15 | Krt7 | 12.78 | 4 | 2 |
| 16 | Batf3 | 11.65 | 1 | 1 |
| 17 | Mbnl3 | 11.44 | 6 | 1 |
| 18 | Klf4 | 11.34 | 9 | 1 |
| 19 | Dppa5a | 11.03 | 4 | 3 |
| 20 | Gpx3 | 10.54 | 8 | 2 |

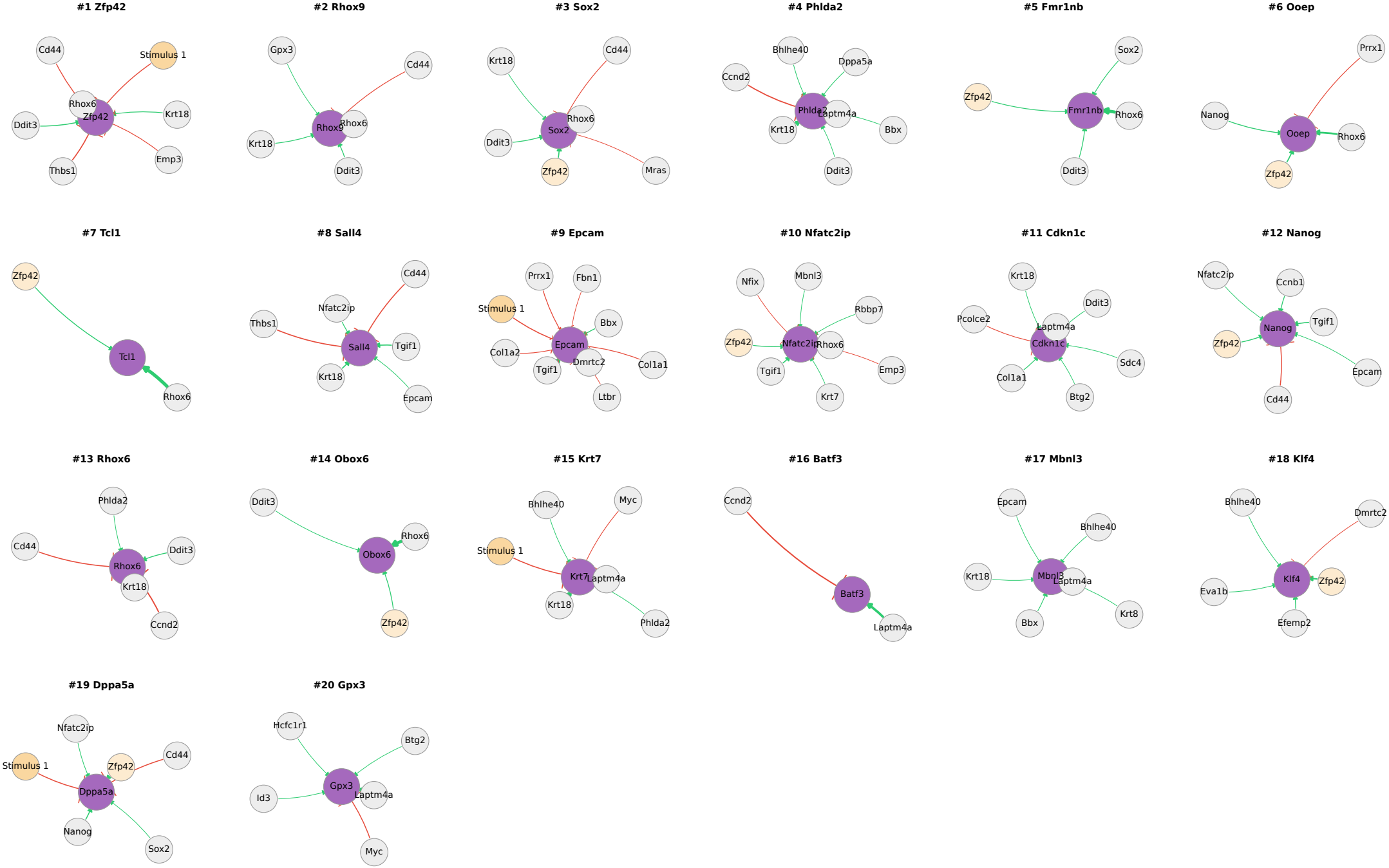
# 2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



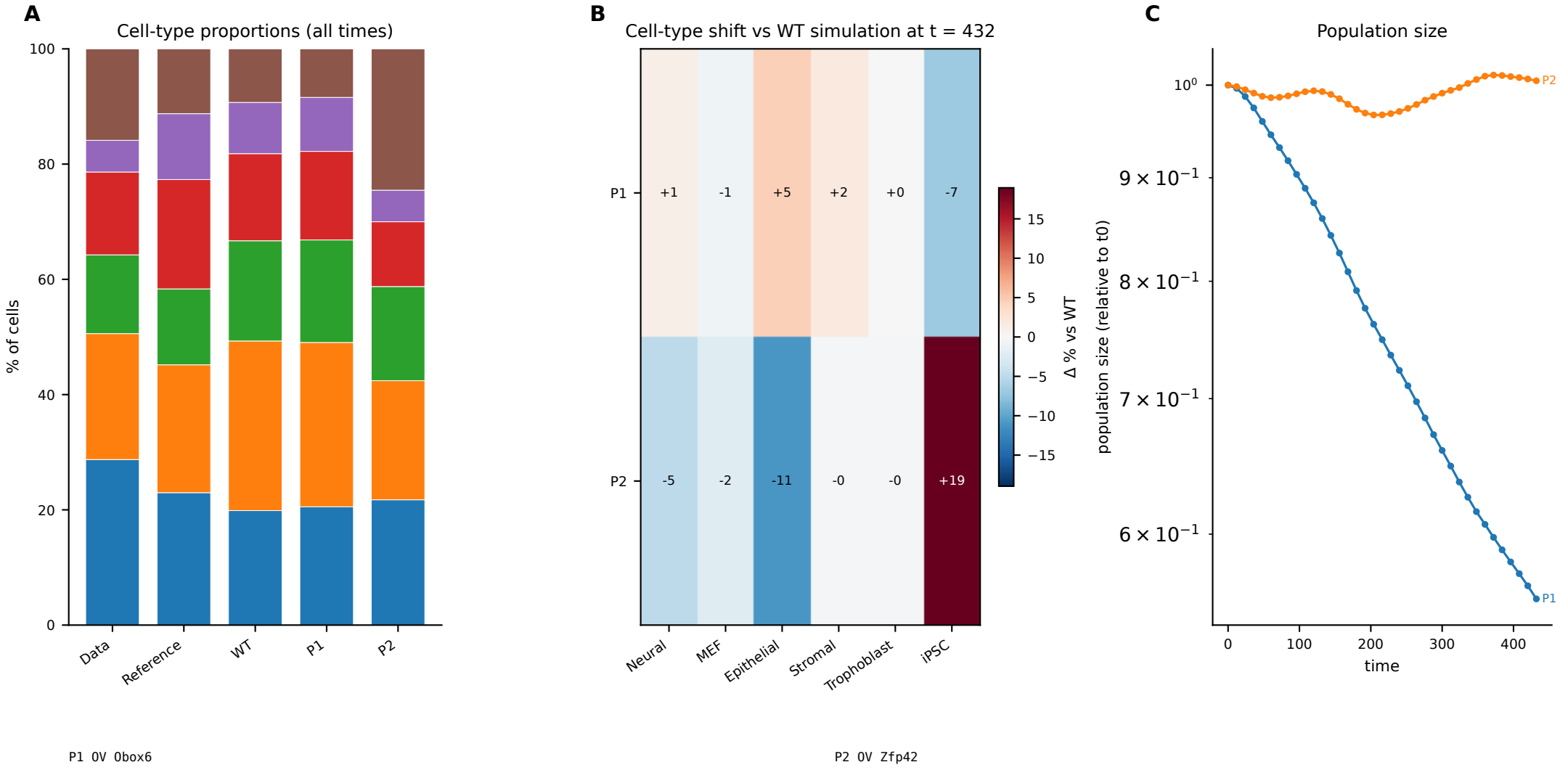
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# 3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

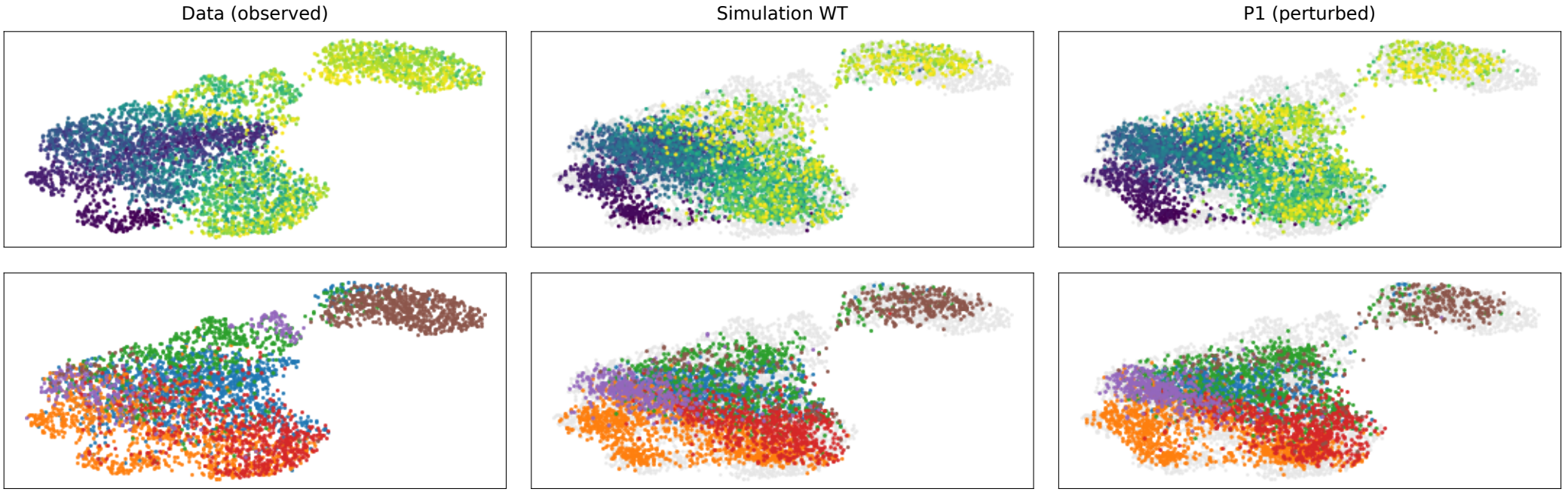


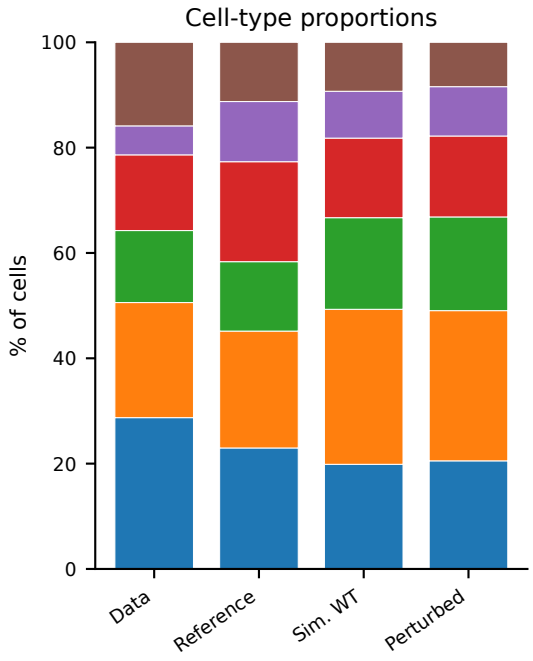

P1 OV Obox6

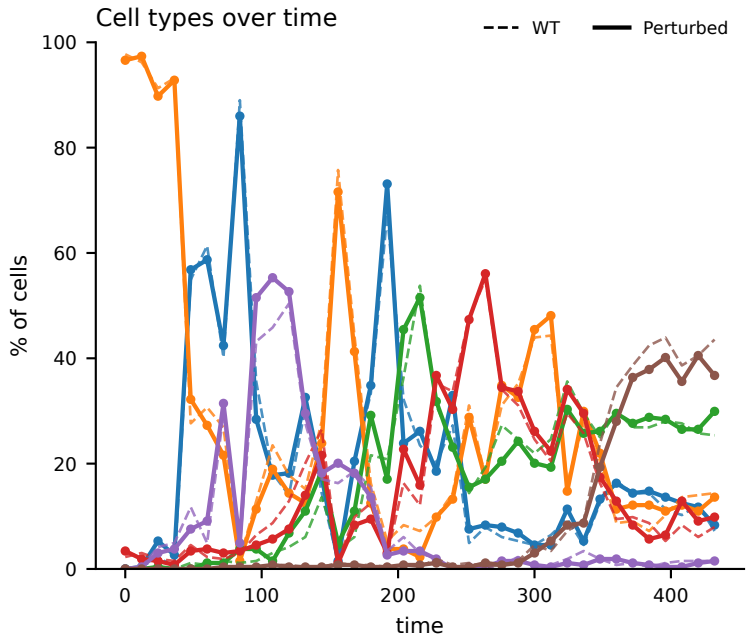
P2 OV Zfp42

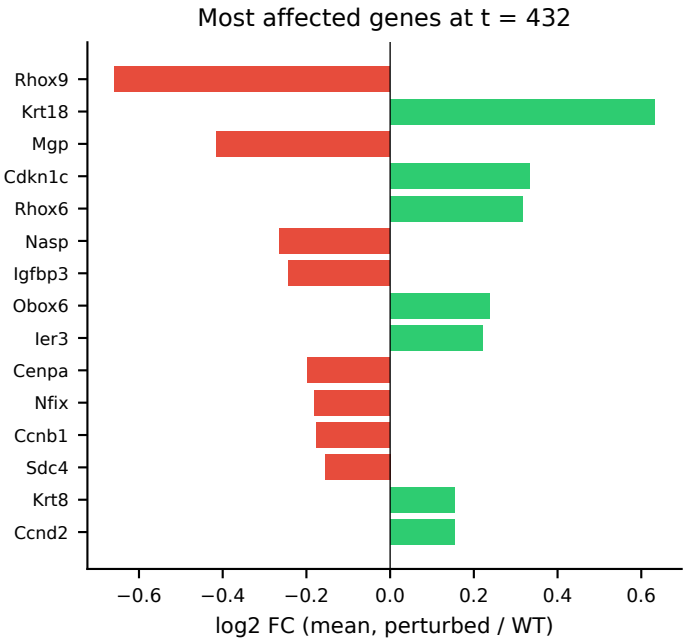
# 3. Perturbation P1 — OV Obox6

KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



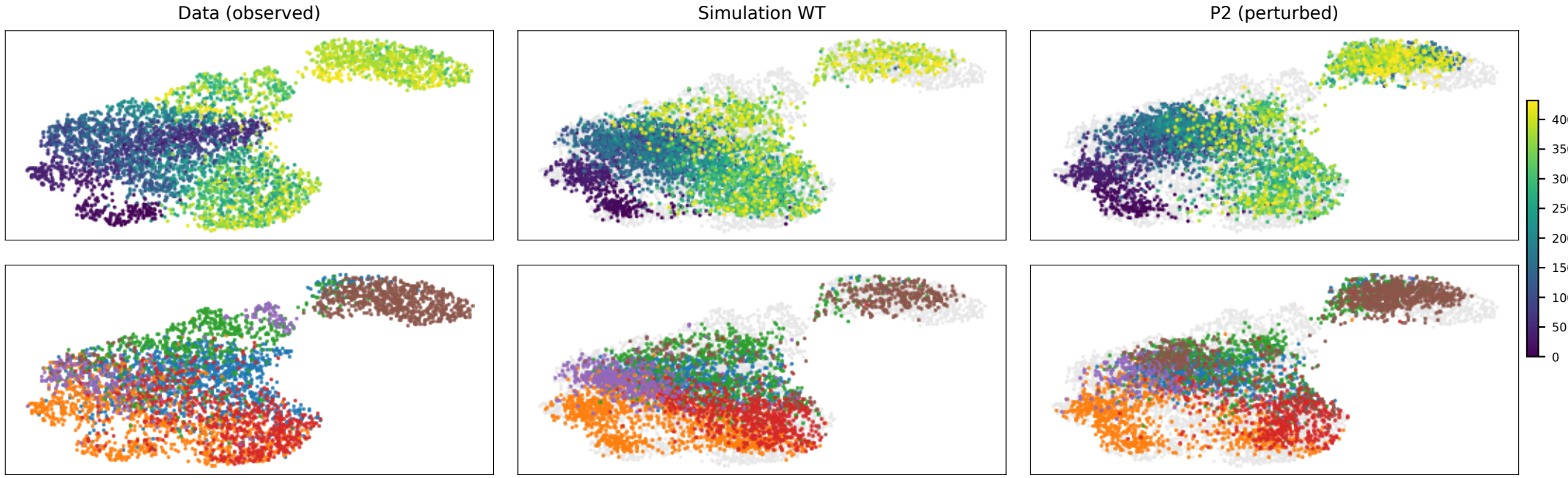



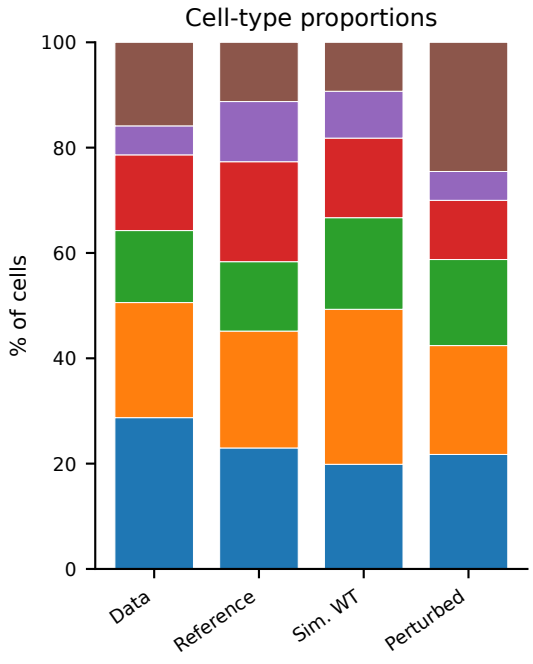



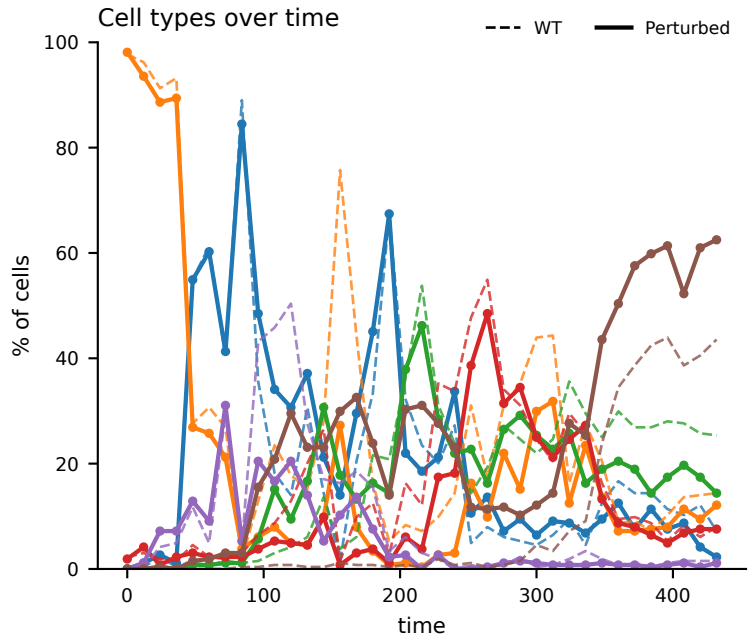

Neural MEF Epithelial Stromal Trophoblast iPSC

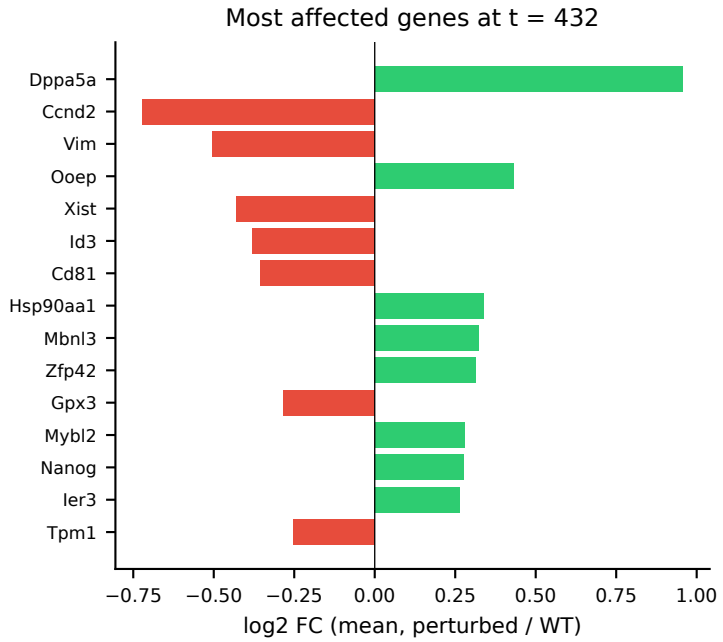
# 3. Perturbation P2 — OV Zfp42

KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).




Neural MEF Epithelial Stromal Trophoblast iPSC

# 4. Proliferation

**Section unavailable:**

```
FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
```

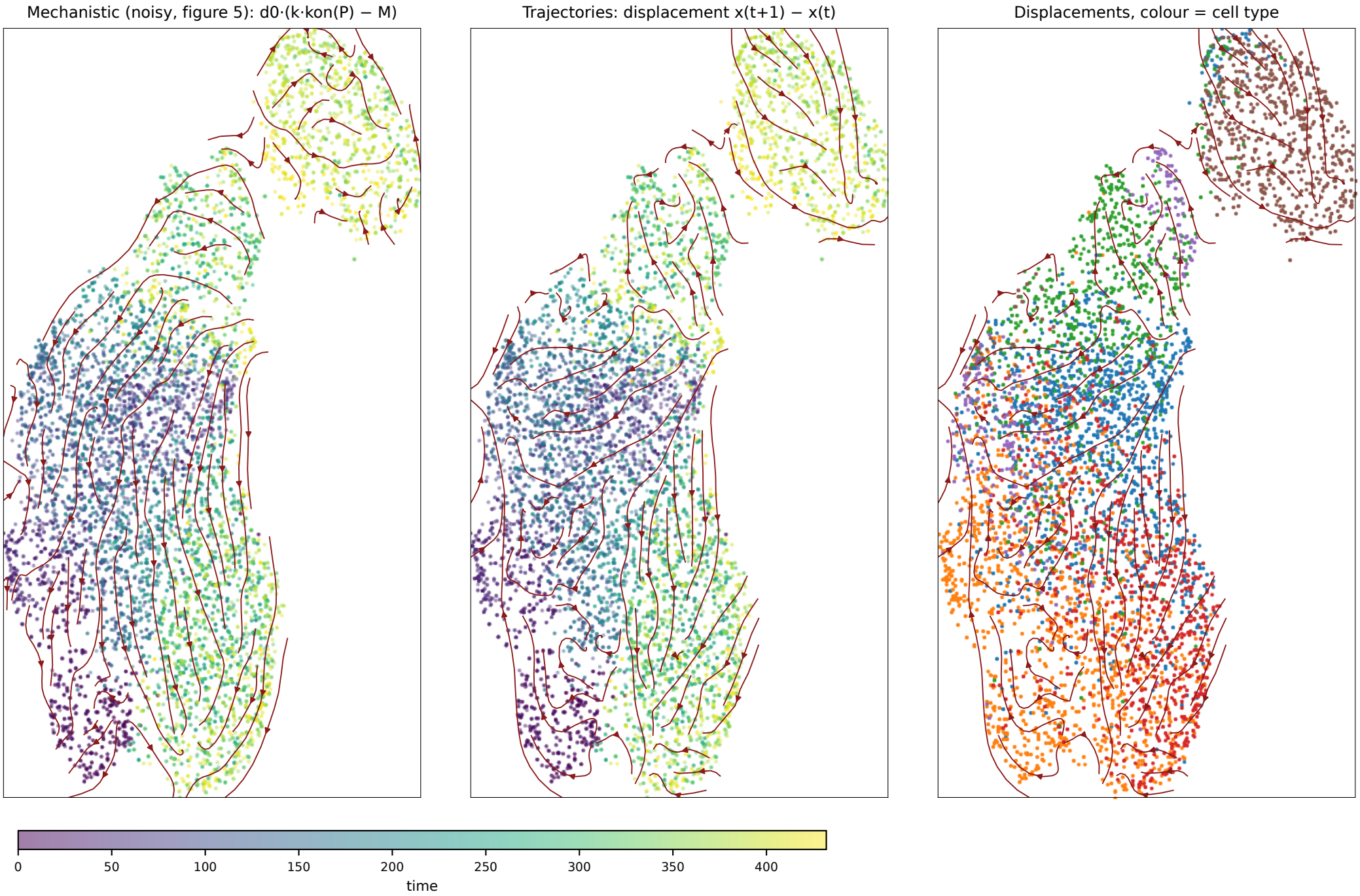
# 4. Protein dilution

**Section unavailable:**

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

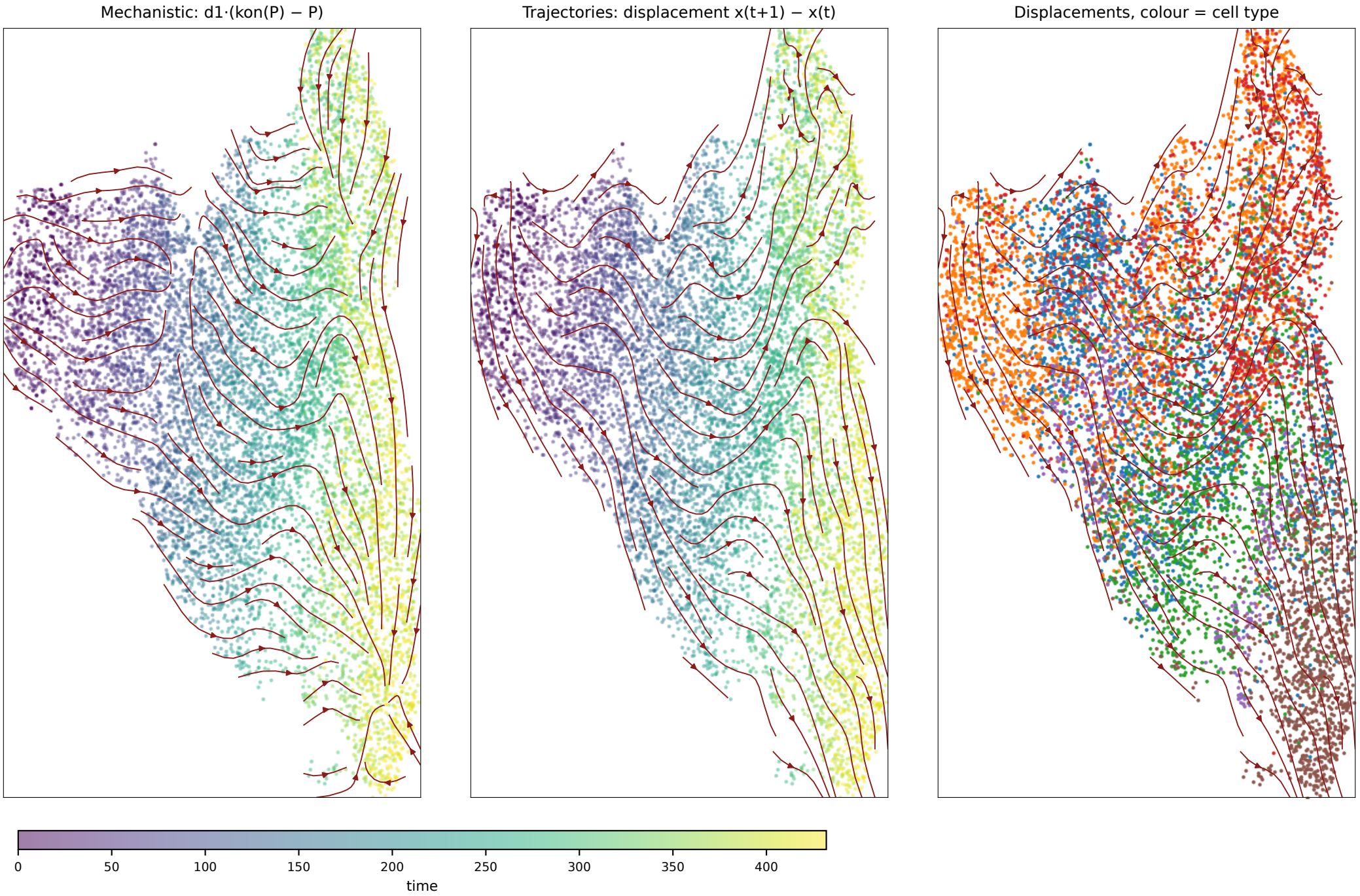
# 5. Learned dynamics — mRNA velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.08.



# 5. Learned dynamics — Proteins velocity and displacement fields

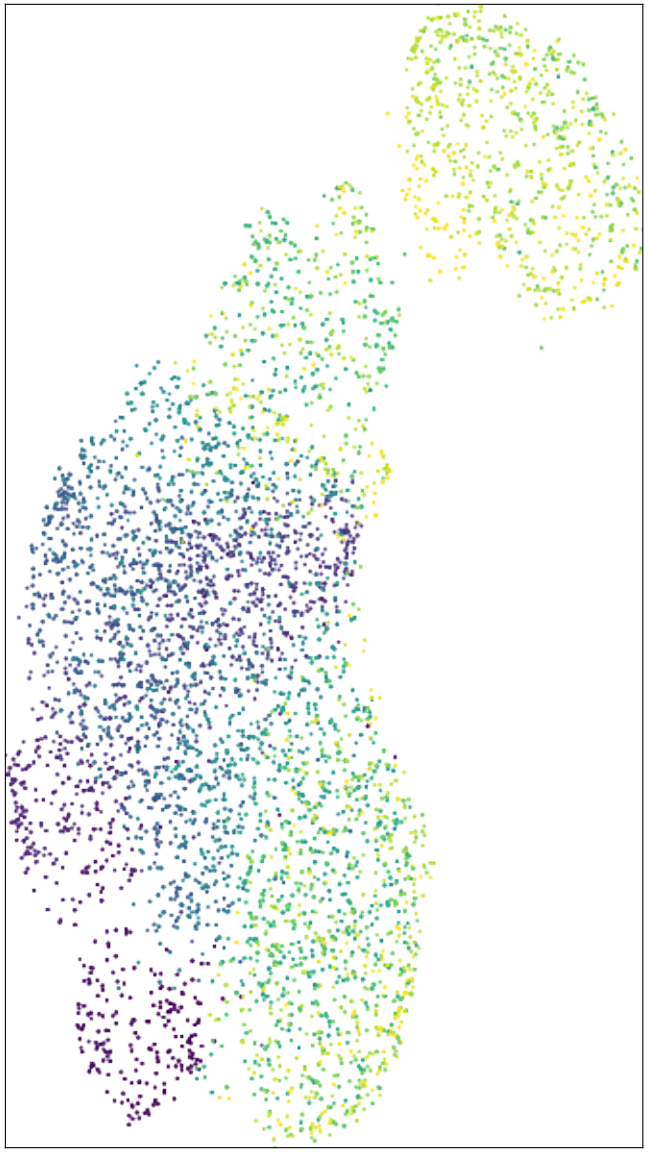
All 9768 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.38.

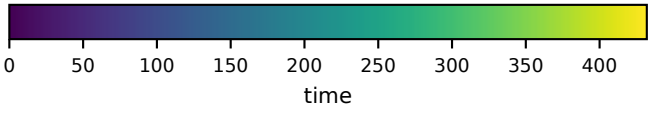


# 5. Learned dynamics — summary

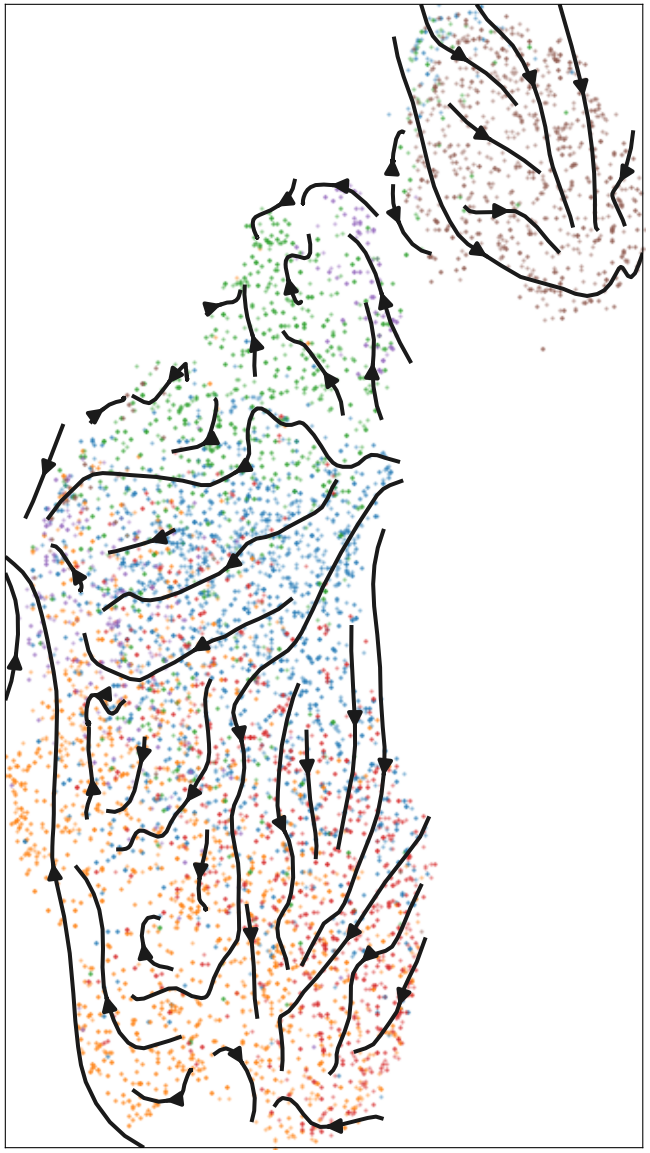
All 9768 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.

Time






mRNA displacement along trajectories, cell_type

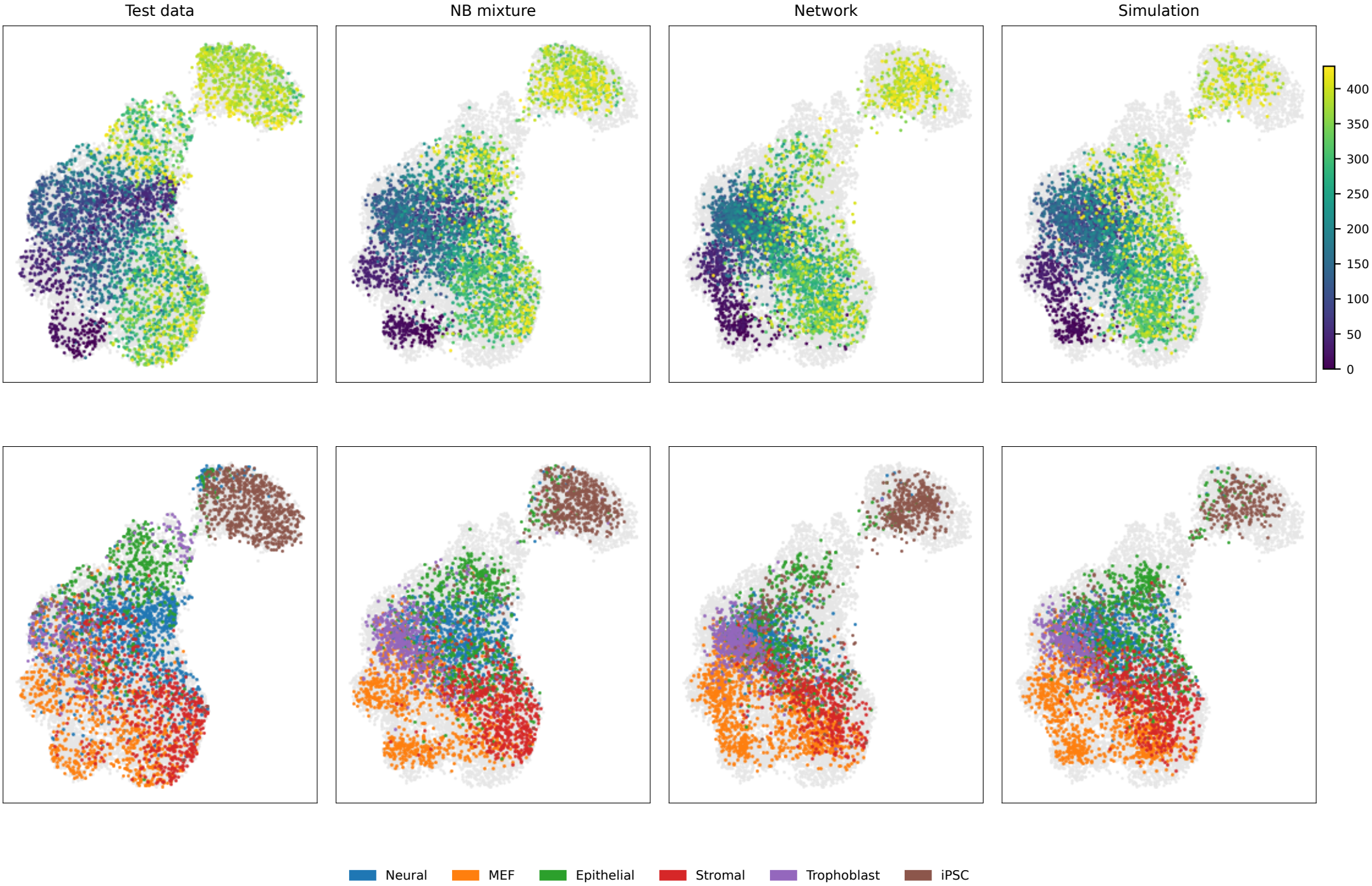


No net proliferation rate



# 6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



# 6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

